

RoboKinoPlatform



500 interactive
Assets



5 Controllers with 3-5
Variations Per Control Task

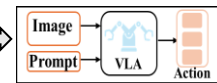


Mainstream Models
for Evaluation

Holistic Pipeline



Data Collection



Model Training



Evaluation



Interpretability Diagnostic

Model A has low task completion
percentage on Task B
Model B outperforms on Task C

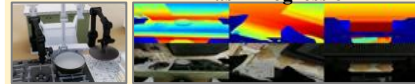
Dual-Arm Scene & Data Collection



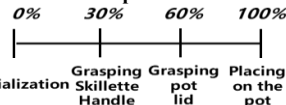
Controller & Tasks



Task Progression



Task Completion Status



Process Evaluation

Spatial

- Environment
- Collisions
- Self Collisions
- Slip Count

Trajectory

- Cartesian Path Length
- Trajectory Length
- Joint Acceleration
- Orientation Path Length

Controller

- Pick/Place
- Push/Pull
- Handover
- Swing/Open/Close
- Rotating

Benchmark Metrics

- Task Completion Time (TCT)
- Success Rate (SR)
- Dangerous Behavior Rate (DBR)
- Task completion percentage (TCP)